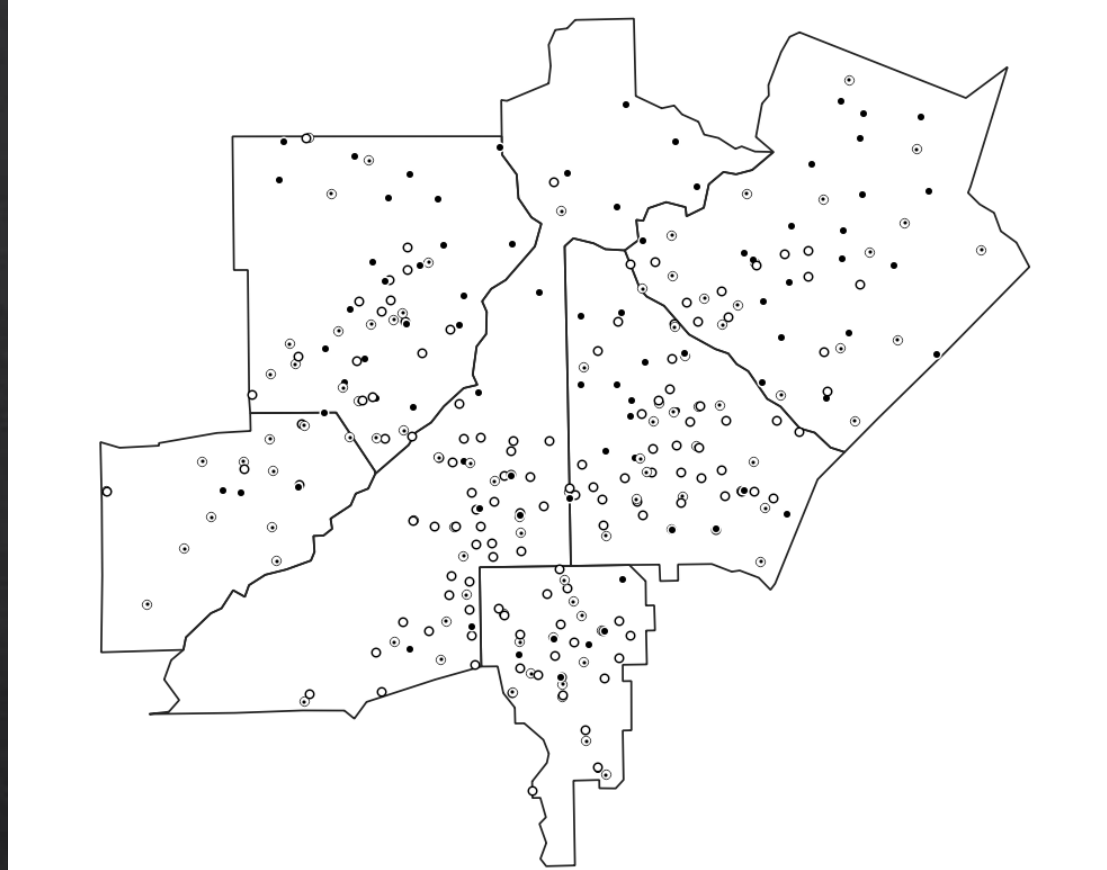




Week 5: Spatial data; Point pattern analysis

Geog4300: Harper

This week: Describing spatial data

Point pattern analysis

Mean center

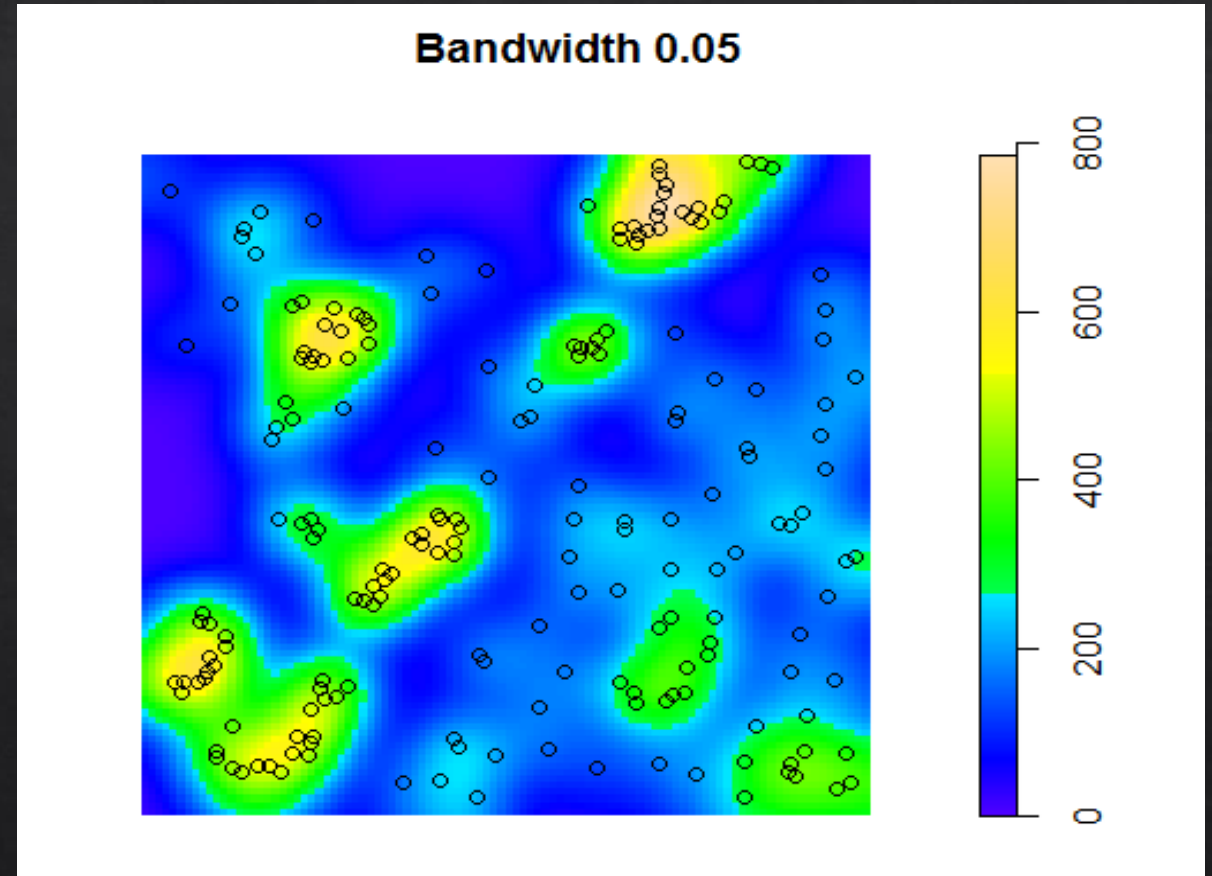
Nearest neighbor index

Kernel density

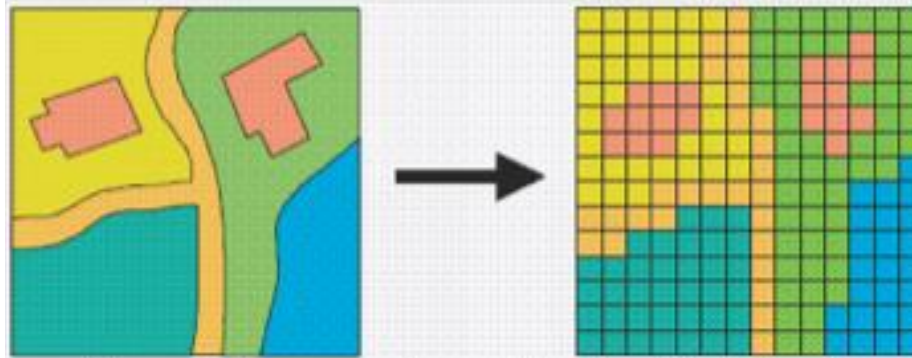
Location quotient

Quadrats

MAUP

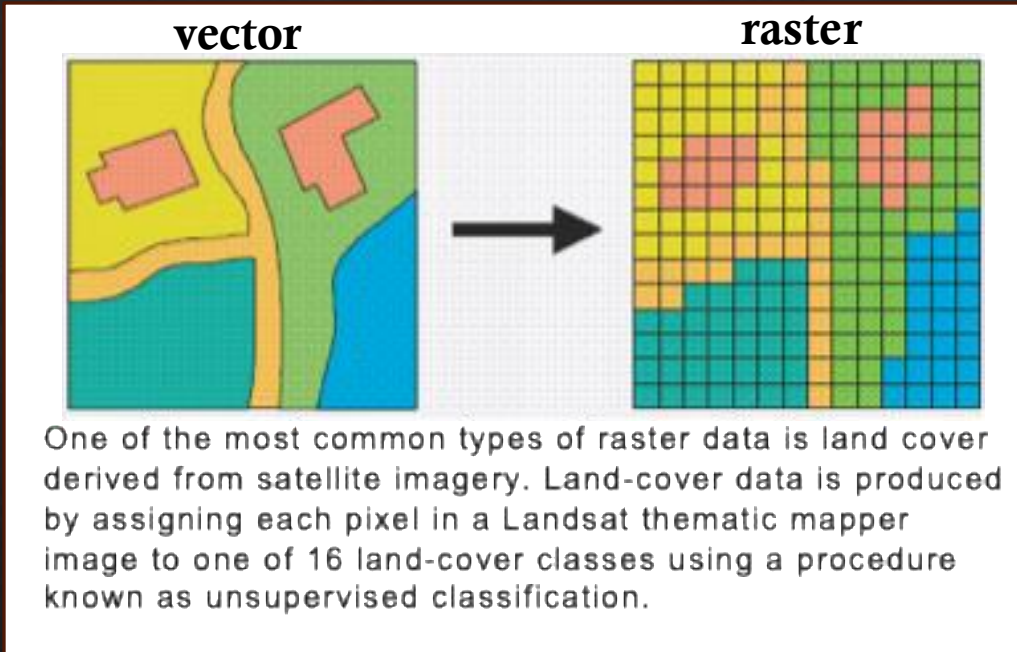


Vector and raster data – which is which?



One of the most common types of raster data is land cover derived from satellite imagery. Land-cover data is produced by assigning each pixel in a Landsat thematic mapper image to one of 16 land-cover classes using a procedure known as unsupervised classification.

Vector and raster data



RASTER DATA:

Comprised of raster cells—all the same size and all with a value

VECTOR DATA:

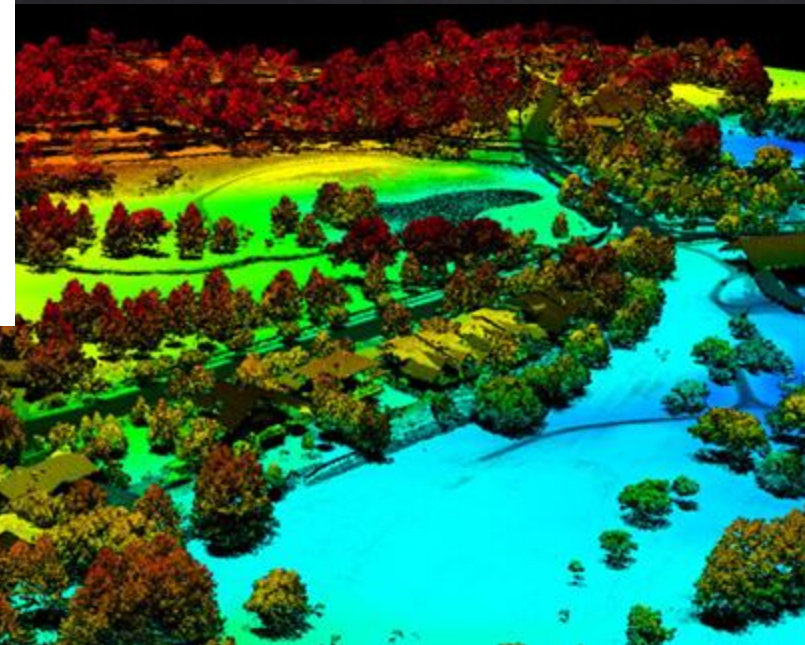
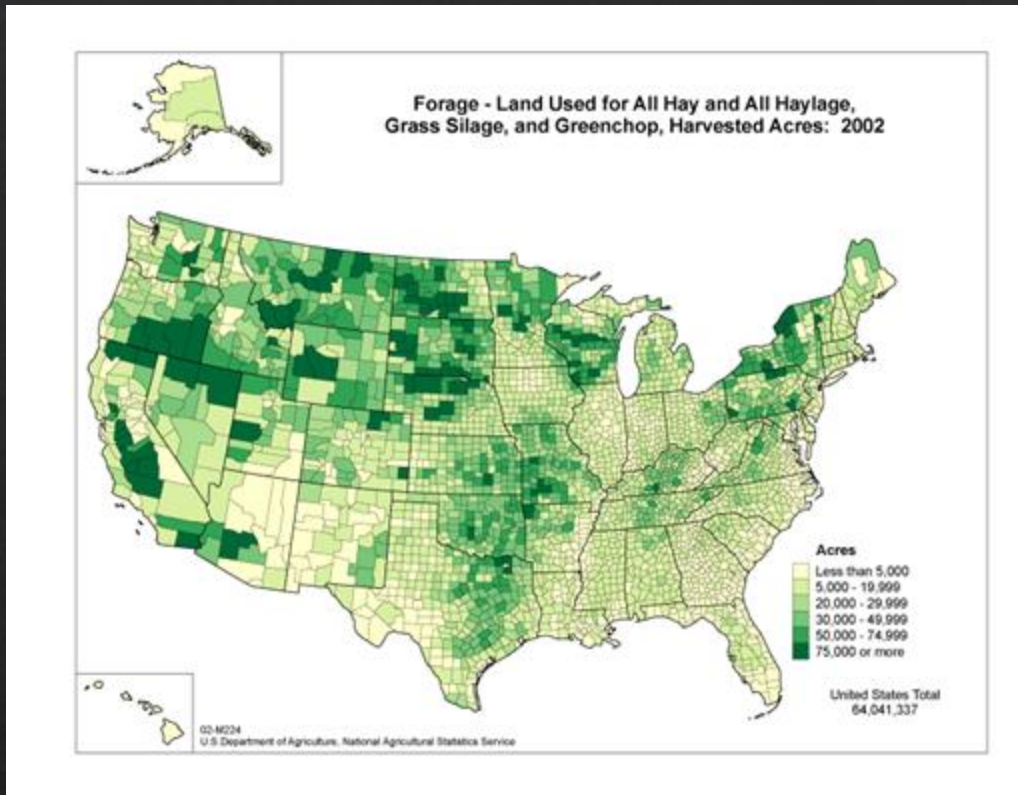
A representation of the world using points, lines, and polygons. Vector models are useful for storing data that has discrete boundaries, such as country borders, land parcels, and streets.

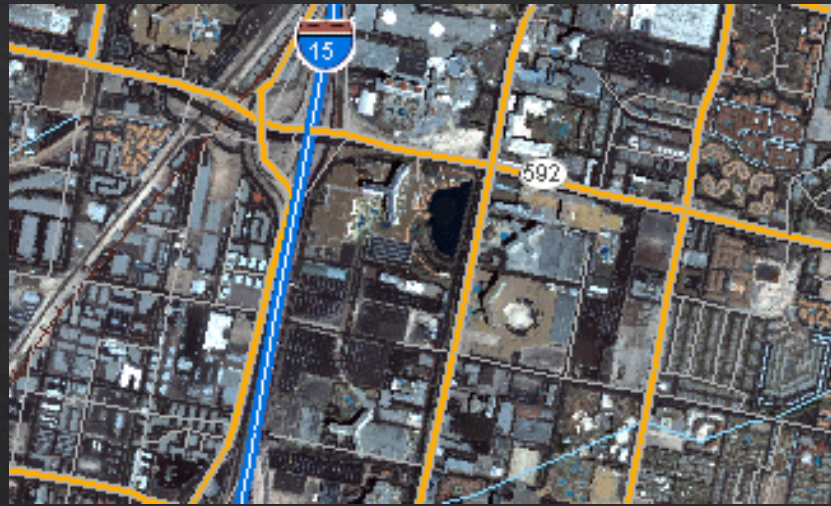
Terminology comes from GIS. I worked with climate data for >10 years before coming across these terms!

Vector vs raster data

- Which is gridded?
- Which is the most appealing when you zoom in?
- Which is the most accurate compared with the basemap imagery?
- How are roads portrayed in vector and raster?
- Which feature type do you think is the most time consuming to produce?
- Which type shows the most detail?
- Give two examples when it would be best to use vector data.
- Give two examples when it would be best to use raster data.

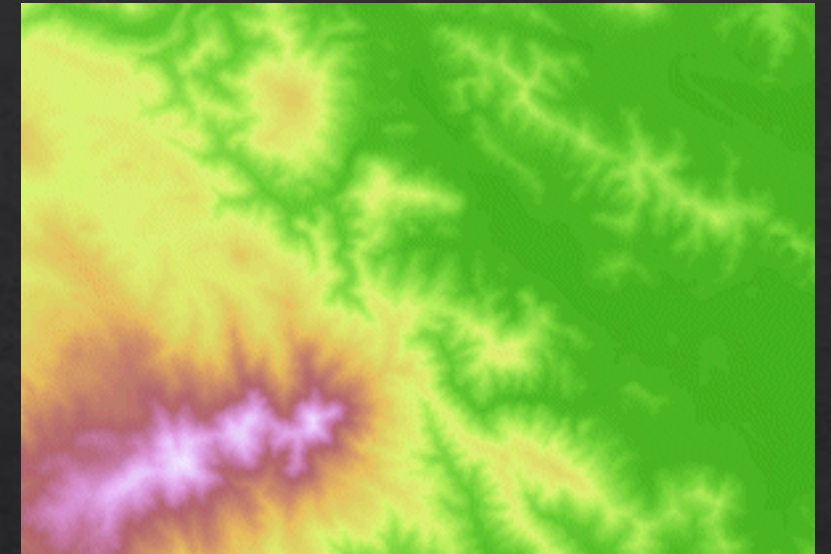
Are these vector or raster?



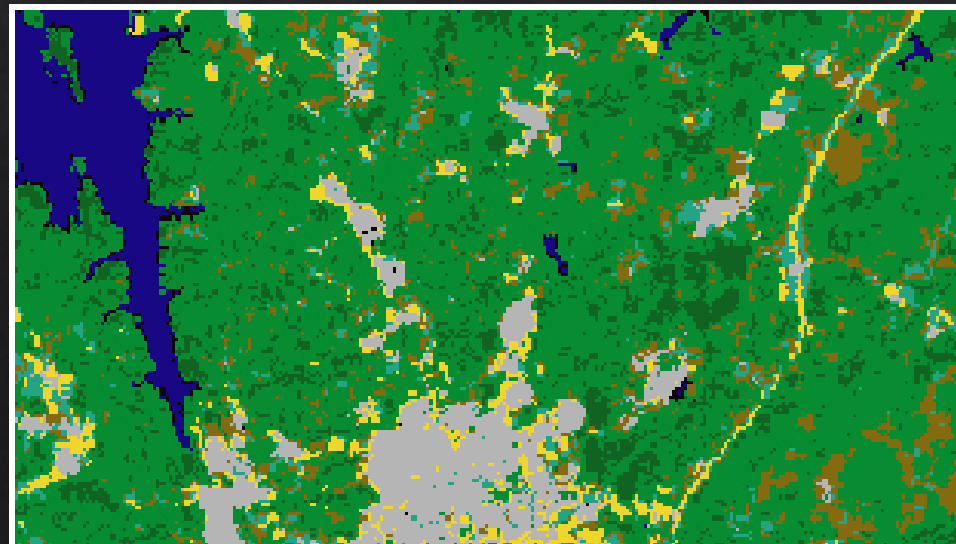


Basemaps

Continuous data

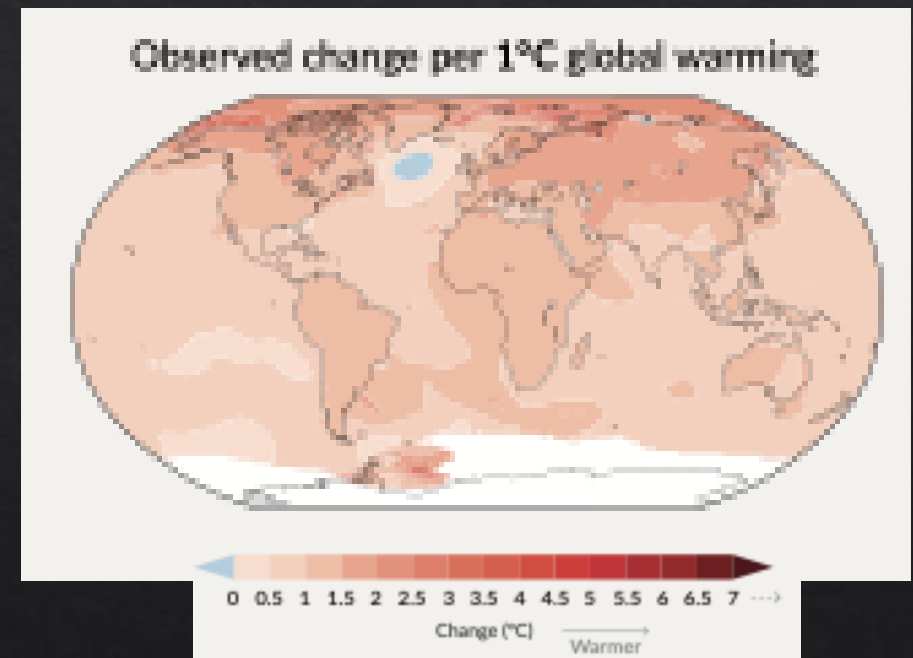
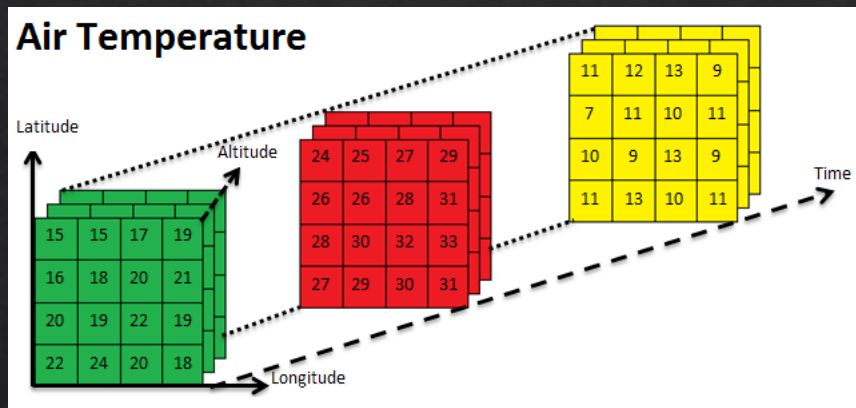


Thematic (discrete) data



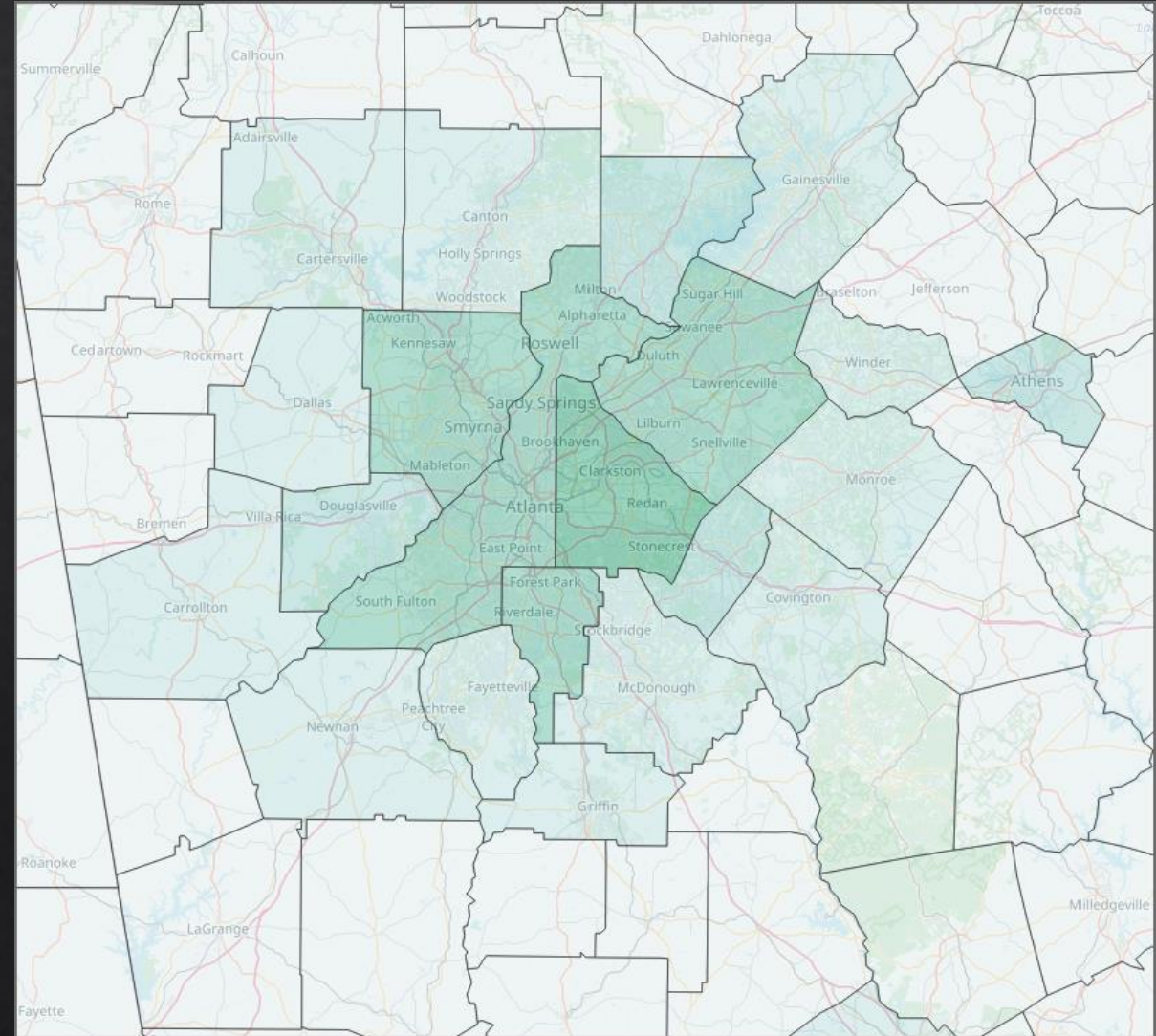
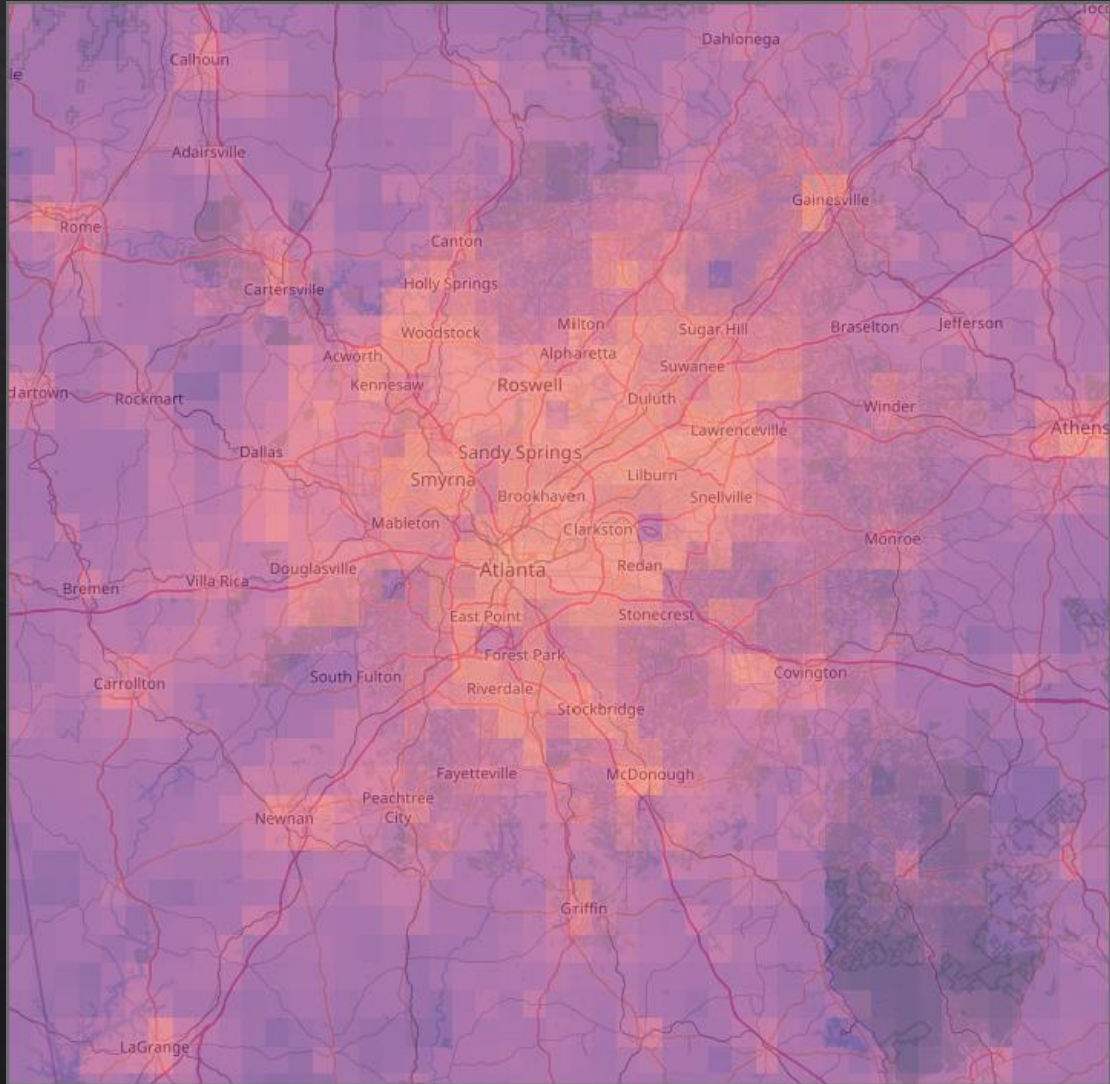
- | | |
|--|---|
|  Agriculture |  Grass |
|  Bare ground |  Pine |
|  Water |  Shadow |
|  Deciduous |  Urban/Developed |
|  Deciduous/
Pine mixed | |

Arrays (multi-dimensional)



Spatial data 101: Vector and raster data

Two maps of population density



Spatial data 101: File types

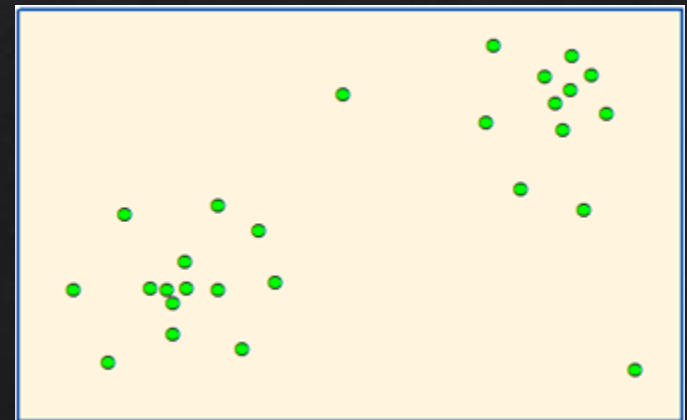
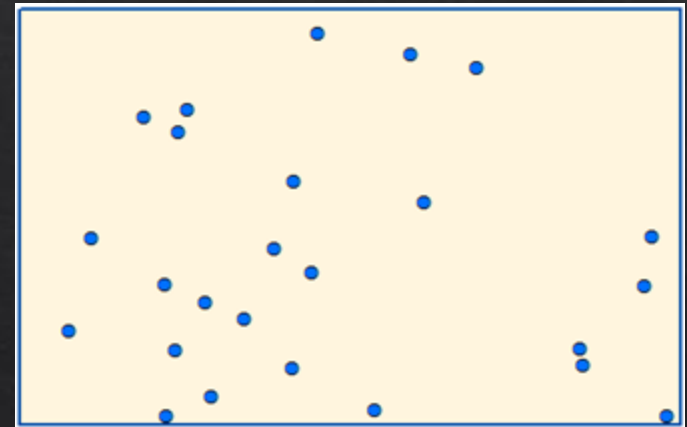
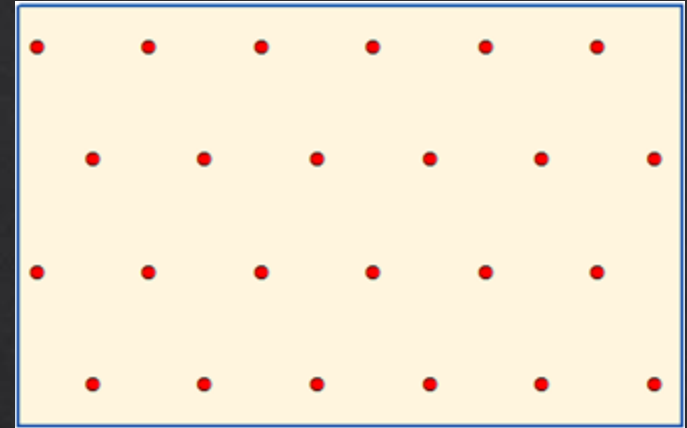
1. Vector: Shapefile, GeoJSON, KML, Geopackage, CSV, XLX,...

2. Raster: GeoTIFF, NetCDF, etc.

Point pattern analysis

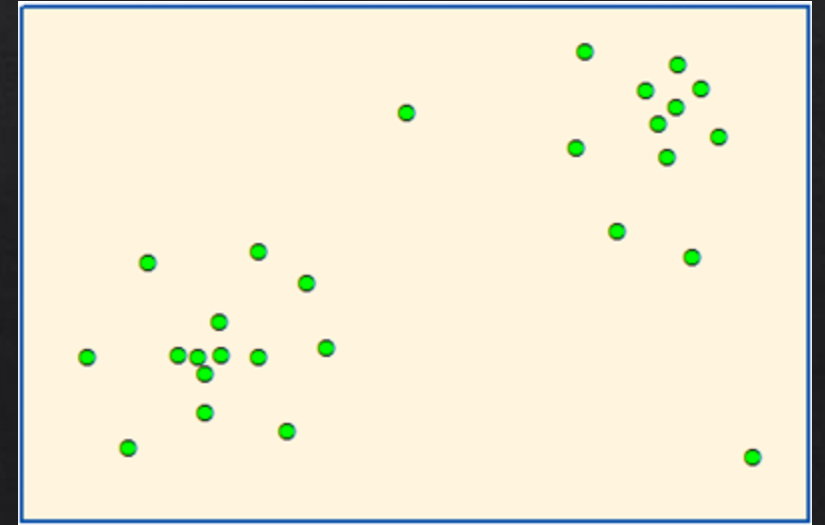
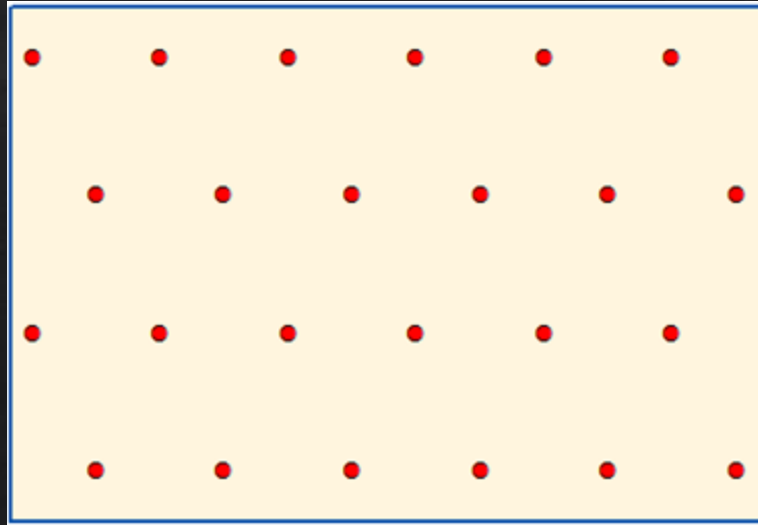
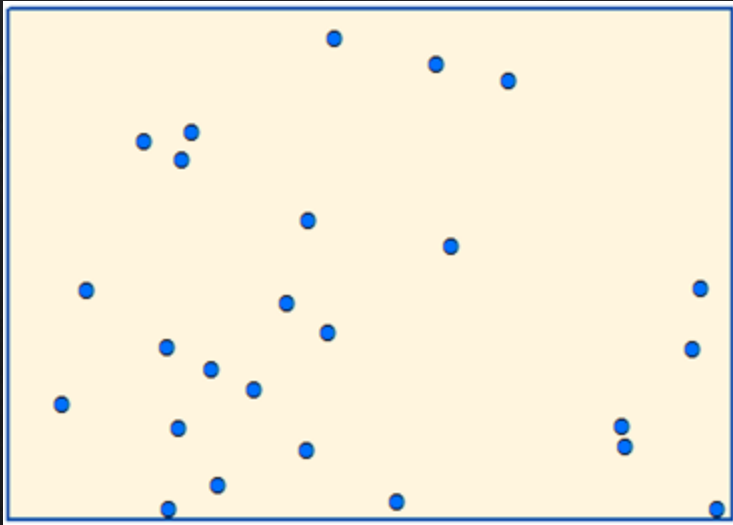
How do we identify patterns in the distribution of point data?

- Clustering and dispersion of point data
- Distribution of point incidents (disease, natural disasters, species, crime, etc.)
- Relative frequency of subevents (strong tornadoes, particular disease variants, etc.)



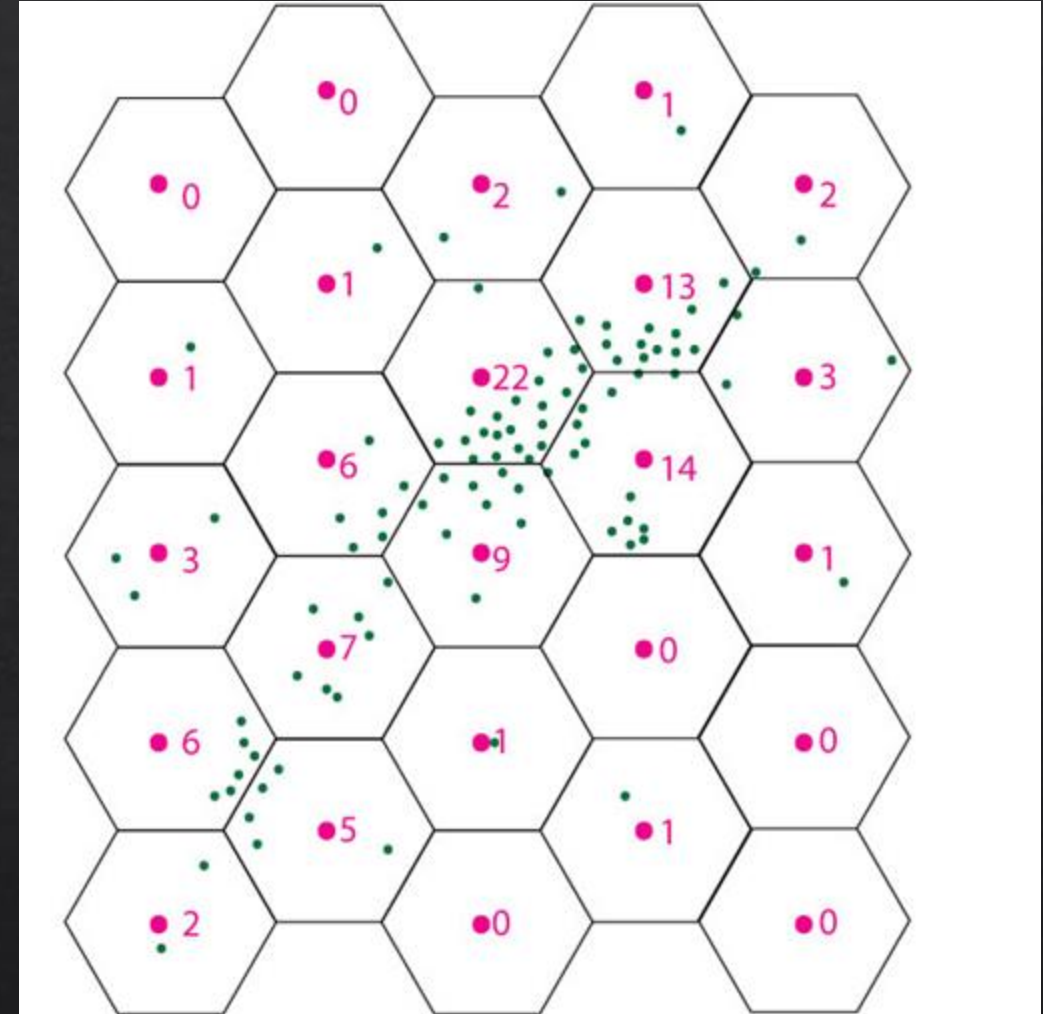
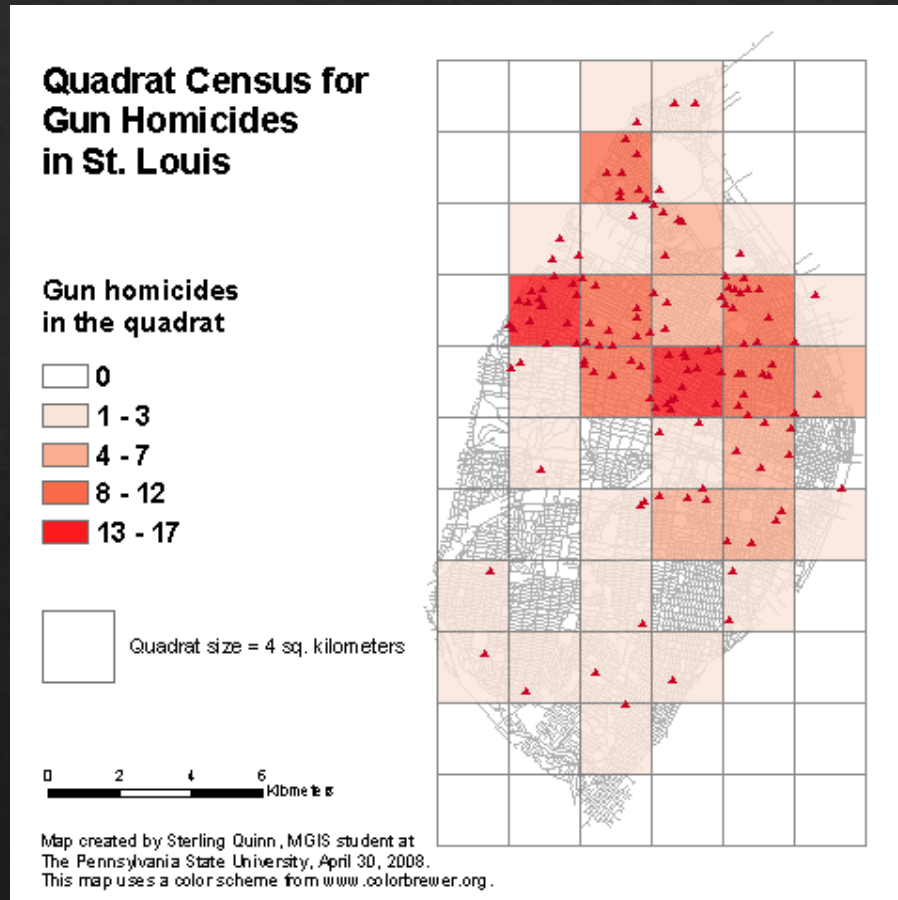
Nearest Neighbor Index

$$\frac{\text{Actual average distance}}{\text{Distance for random distribution}}$$



Quadrats

Aggregating points to a tessalated grid, usually squares or hexagons



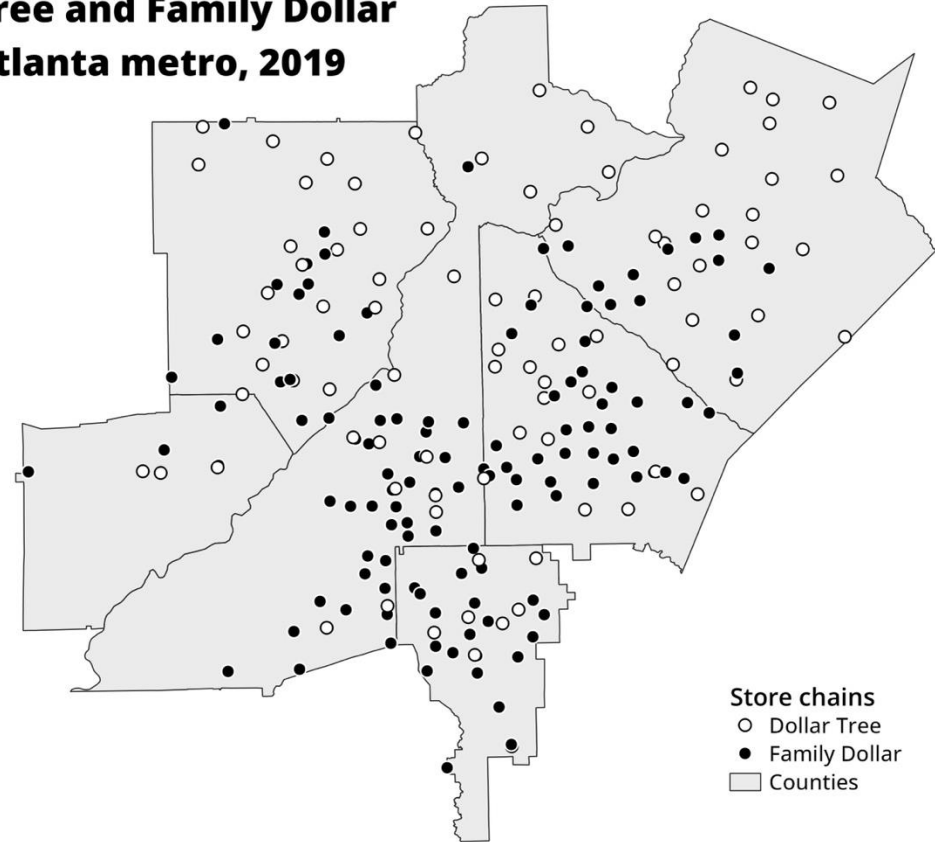
Let's try it!

Dollar stores in Atlanta

Create a quadrat grid of 1.5 or 3 inch squares

Count the total stores for each chain in each grid cell

**Dollar Tree and Family Dollar
in the Atlanta metro, 2019**



Location quotient

Relative incidence of a particular event

Pick a chain and count just those points

What is the percentage of all stores for each grid cell?

$$LQ = \frac{pct_{subarea}}{pct_{area}}$$

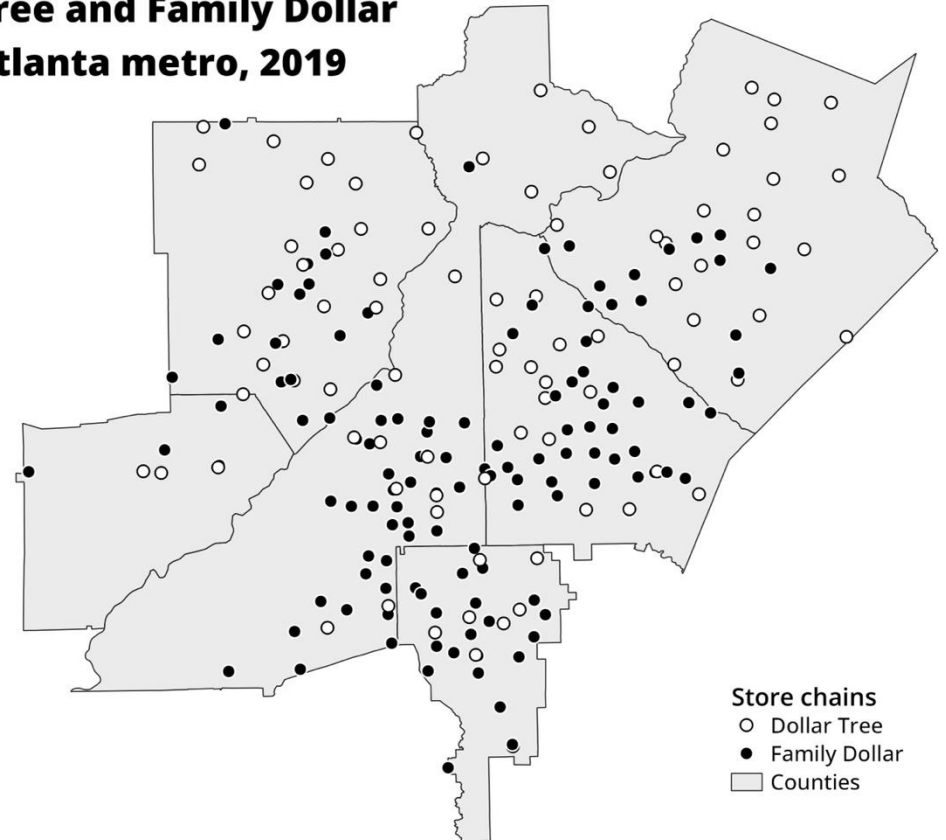
For the whole area:

Dollar General: 31% (93/305)

Dollar Tree: 28% (86/305)

Family Dollar: 41% (126/305)

**Dollar Tree and Family Dollar
in the Atlanta metro, 2019**



The Modifiable Areal Unit Problem

How did our quadrat maps vary?

200	100	400
200	300	300
500	400	100

Base Population

10	20	10
50	10	5
60	10	20

University degree (count)

5 %	20 %	3 %
25 %	3 %	2 %
12 %	3 %	20 %

a - scale effect

6 %
8 %
9 %

10 %	
	7 %
8 %	

b - zoning effect

	4 %	
16 %		6 %